

The Gradient Represents the Directional Derivative Through the Metric

The Problem

At a point p of a manifold, the derivative of a function is a covector:

$$df_p : T_p M \rightarrow \mathbb{R}.$$

It is a linear functional. Given a tangent vector X_p , it returns the number

$$df_p(X_p),$$

the directional derivative of f in the direction X_p .

But gradient descent wants a vector. So we ask:

Which tangent vector contains the same information as the covector df_p ?

Representation by the Metric

If g is a Riemannian metric, then every tangent vector Y_p defines a covector by

$$X_p \mapsto g_p(Y_p, X_p).$$

So vectors can represent linear functionals through the metric.

The gradient is defined as the unique vector

$$\text{grad}_g f(p) \in T_p M$$

whose metric pairing with every direction X_p reproduces the directional derivative:

$$df_p(X_p) = g_p(\text{grad}_g f(p), X_p)$$

for all $X_p \in T_p M$.

This is what “represents” means: the covector df_p and the vector $\text{grad}_g f(p)$ give the same numerical answer after the vector is paired with X_p using g_p .

Euclidean Example

In $\mathbb{R}^n$ with the Euclidean dot product,

$$Df(x)[v] = \sum_i \frac{\partial f}{\partial x^i}(x) v^i.$$

The vector

$$\nabla f(x) = \left(\frac{\partial f}{\partial x^1}, \dots, \frac{\partial f}{\partial x^n} \right)$$

satisfies

$$Df(x)[v] = \nabla f(x) \cdot v.$$

So the Euclidean gradient is the vector representing the covector $Df(x)$ through the Euclidean metric.

Non-Euclidean Metric

If the metric has components g_{ij} , the representing vector changes:

$$(\text{grad}_g f)^i = g^{ij} \frac{\partial f}{\partial x^j}.$$

The covector df is the same differential object, but the vector representing it depends on the metric.

Wasserstein Analogy

In Wasserstein geometry, the role of the metric pairing is

$$\langle u, v \rangle_\rho = \int u \cdot v \, d\rho.$$

When we write

$$D\mathcal{F}(\rho)[v] = \langle \text{grad}_{W_2} \mathcal{F}(\rho), v \rangle_\rho,$$

we are using exactly the same idea: the Wasserstein gradient is the velocity field that represents the directional derivative through the Wasserstein metric.